VeriLoom — Round 1 Submission

Team name: THE_KING
Participant: Thunuguntla Venkata Pavan Kumar
Team size: 1 (solo participant)
Event: We Code Coders — Inkloom AI problem statement
Submission round: Round 1 — The Plan

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1. Understanding of the problem statement

Growing organisations create content through many employees, agencies, and AI tools. As more people contribute, it becomes harder to ensure that every piece of content follows the organisation's brand guidelines and uses current, approved facts, terminology, messages, and visual assets.

This can lead to inconsistent tone or wording, outdated or incorrect information, unsupported promotional claims, off-brand visuals, and accessibility issues reaching customers. These problems can confuse people and damage the organisation's credibility and trust.

We understand the challenge as building an intelligent brand-governance platform that reviews content before publication against an organisation's approved guidance and sources. It should clearly explain each issue, show the relevant guideline or source, and suggest a meaningful correction that preserves the creator's intent. Creators and authorised reviewers should stay in control of edits and approvals, with collaboration and revision history. The platform must also protect confidential materials and be transparent about AI and external services.

Supporting evidence: In the 2025 Content Marketing Institute and MarketingProfs enterprise survey, 47% of surveyed enterprise marketers reported workflow or content-approval challenges and 61% reported cross-team communication challenges across organisational silos among the same 310 enterprise-marketer respondents (drawn from 1,186 global responses fielded between 25 June and 16 August 2024; the same survey also reported that 75% of respondents' organisations used generative AI while 62% had generative-AI usage guidelines). In addition, qualitative public-sector guidance from the UK Government Digital Service (GOV.UK content principles) explains the operational reason for shared style rules when many people create content: common conventions give writers and editors consistent answers on style, format, and accuracy so readers encounter coherent material. These sources provide public context on content-governance challenges, but the survey is self-reported and the GOV.UK guidance is qualitative—they do not establish causation, substitute for user interviews, or prove VeriLoom's effectiveness. No customer interviews or product performance measurements have been invented or claimed.

2. Users affected by the problem

User role	Need and difficulty today	Planned role in VeriLoom
Content Creator Employee or agency contributor	Needs to know what to fix before publication without losing the purpose and intent of the draft.	Submits draft content for review, reads cited issues and suggested edits, revises manually or by applying a suggestion, and resubmits.
Reviewer Manager, team lead, or authorised brand reviewer	Needs to verify claims and wording against current rules, ask why an issue was flagged, and record a documented decision.	Reviews the draft and findings, asks grounded follow-up questions, adds comments or suggested changes, and approves or requests revisions.

User role	Need and difficulty today	Planned role in VeriLoom
Brand Administrator Governance owner	Needs one controlled, versioned set of current brand rules, approved facts, terminology, key messages, and assets.	Creates the organisation profile, maintains approved guidance, facts, terminology, messages, and assets with versions, and assigns user roles.

Customers are indirectly affected when inaccurate, inconsistent, or inaccessible material is published. The primary users operating the platform are the three internal roles above. A manager or team lead can act as a Reviewer when assigned that permission; the Creator remains responsible for revisions, and the Reviewer remains responsible for the final approval decision.

3. Proposed solution

The proposed product is a **source-grounded brand review workspace**. A Brand Administrator registers the organisation and supplies approved guidance, facts, terminology, and assets. When a Creator submits a draft, the system compares it against the organisation's current approved guidance and is expected to return structured findings—each with a clear explanation, an exact source/version reference, and a suggested edit that preserves the Creator's intent. If the approved sources do not support a decision, the system explicitly marks the case for human review instead of inventing a rule. The Reviewer can question findings, request revisions, or approve the draft, with all comments, decisions, and draft revisions preserved in an audit trail.

The distinguishing workflow is the traceable chain linking **approved source version → finding → reviewer question → creator revision → human decision**. This is designed to let a team inspect why a flag appeared, which rule version applied, and what changed across revisions. AI assists by retrieving evidence, explaining issues, and suggesting edits; it never publishes content autonomously or makes the final approval decision.

4. Complete product workflow

1. **Set up the organisation.** The Brand Administrator creates the organisation profile, adds approved guidelines, facts, terminology, key messages, and assets through the React application, and assigns Creator and Reviewer permissions.
2. **Index approved knowledge.** FastAPI stores original uploaded files in a private Cloudflare R2 bucket and saves object keys and file metadata in PostgreSQL. It extracts text from supported files, splits it into short atomic passages, generates local embeddings using `intfloat/multilingual-e5-small` (with the `passage:` prefix), and saves passage text, embeddings, `organisation_id`, document/section metadata, approval status, and version in PostgreSQL with `pgvector`. Only approved, effective passages for the active organisation are eligible for retrieval.
3. **Submit a draft.** The Creator writes or submits supported text content. FastAPI verifies the user's role and organisation, saves the draft and revision in PostgreSQL, and marks it as `processing`. Draft text is treated strictly as review input and is never mixed into the approved guidance index.
4. **Generate a grounded review.** FastAPI searches only approved guidance for that organisation using hybrid retrieval (BM25 lexical search plus local E5 dense vector search fused with Reciprocal Rank Fusion) and sends the draft with a bounded set of relevant source passages to Sarvam AI. Sarvam AI is instructed to return structured issues, explanations, source IDs/versions, and suggested edits, and to state when evidence is insufficient. FastAPI validates and saves the findings and sets the draft state to `ready_for_review`.
5. **Review and ask follow-up questions.** The Reviewer inspects the draft, cited findings, and suggested edits in React. For a follow-up question, FastAPI loads the active organisation, draft revision, selected finding, cited sources, and a short window of recent chat turns from PostgreSQL. It adaptively routes the question across four controlled paths: reusing cited evidence for direct questions, rewriting ambiguous questions before hybrid search, retrieving fresh evidence for new claims/rules, or asking for clarification / abstaining when no approved evidence supports an answer.

6. **Decide and revise.** The Reviewer adds comments, requests changes, or approves the draft. Comments and decisions are recorded against the draft revision. The Creator sees the feedback, revises manually or applies a suggestion, and resubmits a new revision for review. The Reviewer makes the final approval decision.
7. **Learn from repeated issues.** Later in the build, saved finding categories are expected to summarise recurring compliance issues and highlight brand guidance that may need clarification.

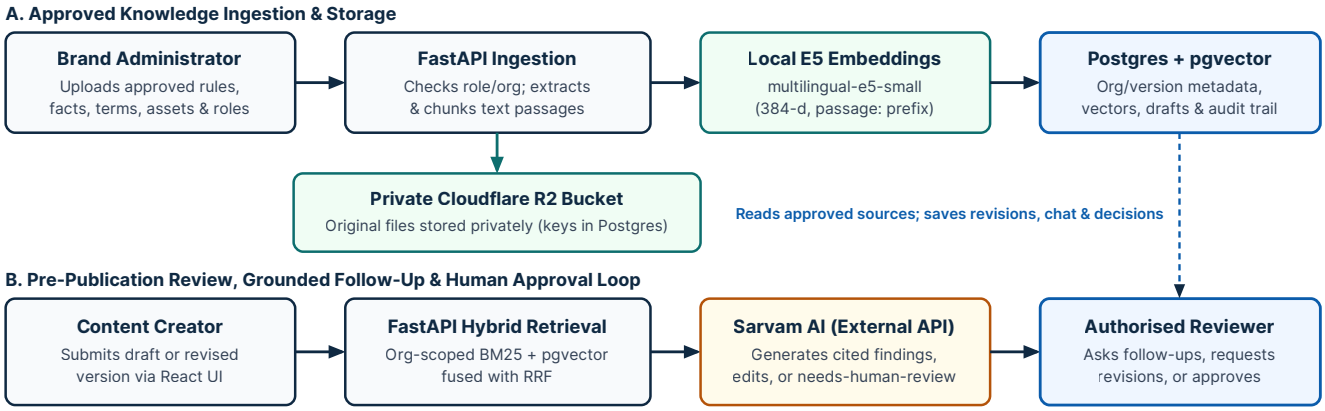


Figure 1. End-to-end brand governance workflow and data placement. Approved brand sources are chunked and embedded locally into PostgreSQL/pgvector while original uploads stay in private Cloudflare R2. Submitted drafts remain separate from the guidance index, and human reviewers retain final approval authority.

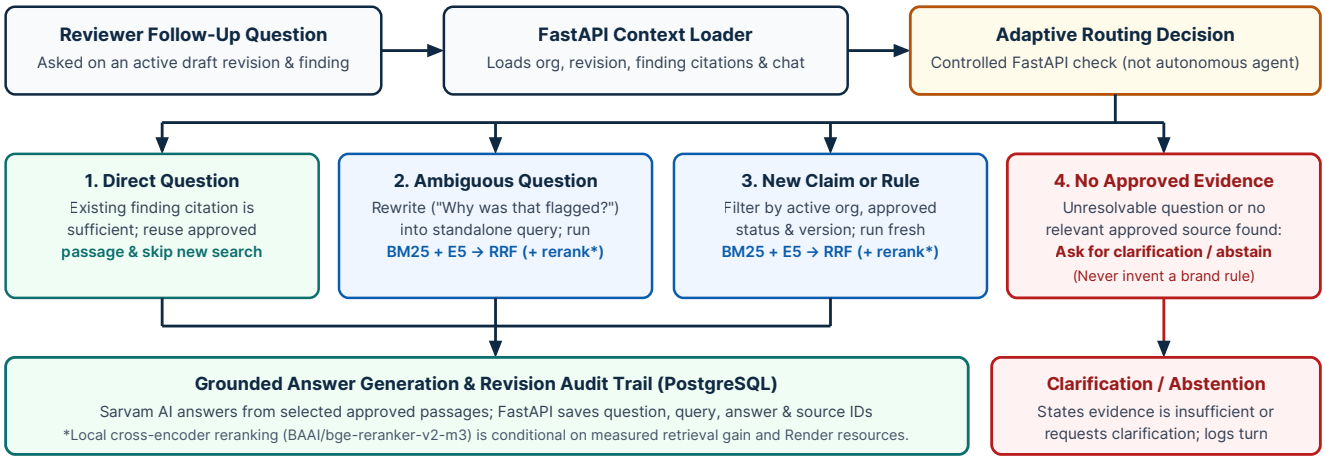


Figure 2. Adaptive reviewer follow-up routing architecture. FastAPI routes reviewer questions across four paths: direct evidence reuse, contextual query rewriting with hybrid retrieval, fresh hybrid retrieval for new claims or rules, and an explicit ask-for-clarification / abstain route when approved sources do not support an answer.

5. Main features intended for the event

Round 2 core MVP: one organisation profile with role-based access (`Creator`, `Reviewer`, `Brand Administrator`); text-based approved-source ingestion with version and status metadata; text draft submission and revision tracking; source-cited findings and intent-preserving suggested edits; reviewer follow-up chat; comments, revision requests, resubmission, and approval history; and a deployed, judge-accessible demo.

Three named improvements tracked across Round 2 and Round 3: The same three named improvements—**Grounded hybrid review**, **Context-aware reviewer follow-ups**, and **Revision-aware human approval**—are used consistently across the architecture, build plan, and evaluation:

Named improvement	Round 2 demonstrable state	Round 3 completed and evaluated state
1. Grounded hybrid review	Retrieve approved, organisation-scoped guidance using BM25 lexical search and `pgvector` semantic search, fuse results with Reciprocal Rank Fusion (RRF), and show a source ID/version beside each finding. Show an explicit `needs human review` result when evidence is missing.	Evaluate retrieval and findings on the development and untouched holdout draft cases; fix missed rules and unsupported flags. Add local cross-encoder reranking (`BAAI/bge-reranker-v2-m3`) only if measured quality improves enough for the Render latency and memory cost. Show actual results and one failure case.
2. Context-aware reviewer follow-ups	Reviewer asks a question about a finding; the application saves the question and answer against the active draft revision in PostgreSQL and cites the approved evidence used.	Route direct questions to existing evidence, rewrite ambiguous questions, run fresh hybrid retrieval for new topics, and ask for clarification or abstain when sources are insufficient. Evaluate route choice, cited source IDs, and answer support on follow-up cases.
3. Revision-aware human approval	Creator submits, Reviewer requests changes, Creator resubmits, and Reviewer approves; each decision and comment records an actor, timestamp, and draft revision.	Complete role and organisation access checks, show a readable revision timeline, prevent stale-revision approval, and summarise recurring finding categories from saved reviews. Verify the complete cycle with separate Creator and Reviewer accounts.

Conditional extra scope and future scope boundary: The primary event workflow focuses on reliable text review. Only if the core text workflow and all three named improvements are stable within the build window will I add one conditional extra input format—such as a text-extracted PDF or a structured social post with image/alt-text metadata fields—in Round 3. Full PDF/image pixel extraction, visual logo comparison against approved assets, and broader governance analytics are placed in conditional extra scope or post-event future scope; they are not treated as the third named improvement. Actual visual compliance checking requires image processing and approved asset comparison, which metadata-only fixtures do not prove.

6. Technology stack and technical plan

Layer	Selected technology	Planned role and deployment fit
Frontend	React, Vite, Tailwind CSS (deployed on Vercel)	Responsive web interface for organisation setup, guidance management, draft submission, cited findings, reviewer follow-up chat, comments, revision timeline, and approvals.
Backend API & workflow	Python, FastAPI (deployed on Render)	Enforces authentication, roles, and organisation access; extracts and chunks text; orchestrates hybrid retrieval, adaptive follow-up routing, and Sarvam AI calls; manages review states.
Database & vector search	Render PostgreSQL with `pgvector`	Stores organisations, users, drafts, revisions, findings, chat history, source metadata, and 384-dimensional embeddings in one managed relational and vector database.
Cloud file storage	Private Cloudflare R2 bucket	Stores original uploaded documents and brand image assets privately; PostgreSQL stores object keys and file metadata.

Layer	Selected technology	Planned role and deployment fit
Language model	Sarvam AI LLM API	Generates structured findings, suggested edits, and grounded follow-up answers from bounded evidence prompts. The API key stays strictly on the backend.
Embeddings & hybrid retrieval	Local `intfloat/multilingual-e5-small` + Python `rank-bm25` + Reciprocal Rank Fusion (RRF)	Creates multilingual semantic embeddings locally inside FastAPI (`query:` and `passage:` prefixes, ≤512 tokens) independent of Sarvam, combined with organisation-scoped BM25 lexical search and RRF rank fusion.
Conditional reranking	Local `BAAI/bge-reranker-v2-m3` on a small candidate set	Multilingual cross-encoder scoring query-passage pairs locally; enabled only if measured retrieval gain justifies added Render memory and latency.
Source control	Git and GitHub	Tracks official build-window progress, evaluation scripts, and reviewable source code.

Deployment fit for local embeddings and BM25: Sarvam's public API catalogue documents chat and other model endpoints, but no hosted text-embeddings endpoint was found in its public documentation. This plan therefore runs `intfloat/multilingual-e5-small` locally inside FastAPI for embeddings and uses Sarvam strictly for LLM generation. Similarly, Render's managed PostgreSQL supports `pgvector`, but its documentation states that ParadeDB's `pg_search` BM25 extension cannot currently be installed directly into Render Postgres (and running a separate ParadeDB service requires paid resources). Using Python `rank-bm25` over the active organisation's approved passages loaded from PostgreSQL avoids a second database service for the solo MVP.

Hardest technical challenge: Returning accurate findings when approved sources contain similar wording, updated facts, or coverage gaps. The plan filters every query by `organisation_id`, `approved` status, and effective version before retrieval; combines BM25 lexical matching with dense vector search; carries canonical document-qualified source IDs (e.g., `NK-FACTS-001#FACT-06`) into every finding; validates structured model output; and enforces an explicit insufficient-evidence / abstention path. Reviewer follow-ups are tied to stored PostgreSQL revision history rather than relying on untracked LLM memory.

Privacy, access control, and external provider disclosure: Role and organisation checks apply to every draft and source query; draft content is never mixed into the approved guidance index; R2 objects are private by default; and API keys remain server-side in Render/Vercel environment settings. Draft excerpts, reviewer questions, and selected source passages sent to Sarvam AI cross the application's database boundary and are processed by an external provider (alongside hosting providers Vercel, Render, managed PostgreSQL, and Cloudflare R2 under their service terms). This external processing is explicitly disclosed; real confidential material should only be used after an organisation approves that arrangement and reviews current provider data-handling terms. Human reviewers retain final approval authority.

7. Responsibilities of each team member

Participant	Planned responsibilities	Contribution evidence
Thunuguntla Venkata Pavan Kumar Sole participant	Problem research, UX design, React/Tailwind frontend, FastAPI backend, PostgreSQL + `pgvector` schema, hybrid RAG and Sarvam AI integration, evaluation runner, deployment, documentation, and demo.	Build-window Git commit history, progress log, deployed working demo, development and holdout evaluation outputs, and individual submission.

There are no other team members. I will complete my own required individual submission and use the same exact team name and legal name across every event form.

8. Development plan across all three rounds

The organiser's official dates control the schedule. **Product implementation must occur within the official build window** to count toward judging. Prior problem research, this Round 1 architectural plan, and the fictional Nook & Kettle data pack are disclosed as pre-build preparation; they are not claimed as judged product implementation.

Round	Planned work and deliverables	Verification and exit evidence
Round 1 — The Plan	• Define the problem statement and affected user roles. • Research the enterprise content-approval challenge. • Specify the end-to-end workflow, hybrid RAG architecture, three named improvements, risk controls, and solo MVP scope. • Prepare a clearly labelled fictional demo and evaluation data pack.	• This Round 1 submission PDF and editable source document. • Detailed `PRESENTATION.md` product plan and public source links. • Synthetic data pack with explicit provenance and boundary disclosures.
Round 2 — The Build	• **1. Foundation:** During the official build window, initialise Git, build the React/Vite/Tailwind and FastAPI apps, provision Render PostgreSQL (`pgvector`) and private R2, and enforce organisation/role authentication. • **2. Approved knowledge:** Register approved brand sources with version/status metadata and local E5 embeddings. • **3. Creator-to-Reviewer loop:** Implement draft submission, grounded Sarvam review with hybrid BM25 + `pgvector` retrieval, follow-up chat, comments, revision requests, resubmission, and approval. • **4. Deploy:** Deploy to Vercel and Render and run initial development-case checks.	• Judge-accessible GitHub repository and deployed link (or functional walkthrough). • Complete architecture, model, API, data, and privacy disclosures. • Full two-account loop (`Creator` and `Reviewer`) with persisted revision history. • Explicit `needs human review` handling for unsupported claims and recorded development-case results.
Round 3 — The Finish	• **1. Deliver & evaluate the three named improvements:** Compare hybrid retrieval against baseline (retaining local `bge-reranker-v2-m3` reranking only if measured gain justifies Render latency/memory), finish adaptive follow-up routing (rewrite, fresh search, and clarify/abstain), harden stale-revision blocking, and add a recurring-issues summary view. • **2. Reliability & UX:** Test failure paths (empty/unsupported source, changed version, Sarvam timeout/malformed output, duplicate submit, wrong role/org), keyboard navigation, and contrast. • **3. Conditional extra format:** Add one extra input format (e.g., text-extracted PDF or social post metadata) *only* if the core text loop remains stable. • **4. Final submission:** Run untouched holdout evaluation and prepare the final report and demonstration.	• Before/after demonstration and linked commit log for each of the three named improvements. • Pass/fail counts, latency measurements, and failure-case analysis on both development and untouched holdout cases. • End-to-end deployed smoke test verification. • Final documentation with limitations, roadmap, and complete participant details.

9. Expected impact

If implemented as planned, VeriLoom is **expected** to deliver the following outcomes (stated strictly as expected outcomes for a Round 1 plan, with no invented performance measurements):

- **Expected outcome for Content Creators:** Receive specific, source-backed feedback and intent-preserving edit suggestions before publication so compliance issues can be fixed without losing draft intent.
- **Expected outcome for Reviewers:** Expected to spend less time manually searching rules and explaining repeated issues while retaining full context and final human approval authority.
- **Expected outcome for Brand Administrators:** Maintain one controlled set of versioned brand guidance and inspect recurring finding categories to see which rules need clarification.
- **Expected outcome for Customers and Organisations:** Expected to reduce inconsistent wording, outdated facts, or unsupported promotional claims reaching the public.

Planned evaluation measures during the event: Rather than claiming unmeasured gains, I will evaluate the built system on separate development and holdout cases using five concrete measures: (1) **source retrieval** (whether expected approved source IDs appear in top retrieved results while excluding archived or other-organisation records), (2) **citation support** (whether findings and follow-up answers cite supporting passages), (3) **unsupported-question and abstention handling** (how reliably unsupported draft claims are marked for human review and unsupported reviewer questions trigger clarification or abstention), (4) **review latency** (response time for normal draft reviews and follow-up queries on Render), and (5) **completion of the revision/approval loop** (whether separate Creator and Reviewer accounts can complete submission, review, follow-up, revision, and final approval while blocking stale revisions). Any user feedback will be reported only if actually collected from real participants.

10. Future scope after the event

After the core text workflow and the three named improvements are validated, post-event future scope includes: direct image, colour, clear-space, and logo pixel comparison against approved brand assets in R2; deeper automated WCAG accessibility checks; broader document and social formats (multi-page PDFs, slide decks, rich HTML templates); dedicated BM25/hybrid search infrastructure (such as ParadeDB or OpenSearch) for larger corpora; CMS and design-tool integrations; notifications; richer cross-campaign governance analytics; and human-led pilots with real organisations following privacy and provider-term reviews. These are roadmap extensions and are not claimed as completed event features.

11. Complete team and participant details

Required detail	Participant record
Exact desired team name	THE_KING
Participant full legal name for certificate	Thunuguntla Venkata Pavan Kumar
Team size	1 — solo participant
Participant email	iec2025049@iiita.ac.in
Institution	Indian Institute of Information Technology, Allahabad (IIIT Allahabad)
GitHub profile	https://github.com/9553943884pavan-alt

Appendix: References, Data Provenance, and Disclosures

Synthetic data pack and evaluation boundaries: This Round 1 plan and the accompanying *Nook & Kettle* / *Cedar Cup* data pack (prepared `2026-11-03`) were created with significant AI assistance and human review. All Nook & Kettle brand rules, product facts, draft examples, expected findings, and follow-up conversations are **fictional and synthetic**; they are not real customer data, live brand assets, or user research. Within the data

pack, only `knowledge/records.jsonl` (45 atomic records with `ingestible: true` and `status: "approved"`) represents normal approved-knowledge ingestion input. All evaluation files (`evaluation/drafts.jsonl`, `evaluation/followup_cases.jsonl`, `evaluation/scope_cases.jsonl`) and isolation fixtures (`knowledge/isolation_fixtures.jsonl`, `ingestible: false`) are **offline evaluation cases only**—they are **NOT** approved brand knowledge, must never be indexed as guidance or passed as prompt answers, and their holdout split must remain untouched until Round 3. Similarly, the public problem evidence (`public_references/problem_evidence.jsonl`) and public reference summaries (`public_references/references.jsonl`, marked `status: "reference_only"`, `ingestible: false`, `adoption_required: true`) are **NOT** approved brand knowledge or customer research; a Brand Administrator must explicitly review and adopt a rule into an organisation before it applies. All judged product implementation must take place during the official build window.

Reference category	Cited public evidence, reference collection, and technical documentation links
Problem statement evidence (`PROBLEM_EVIDENCE.md`, `ingestible: false`)	• 2025 Content Marketing Institute & MarketingProfs Enterprise Content Marketing Survey (310 enterprise marketers from 1,186 global responses fielded 25 June–16 August 2024: 47% workflow/approval challenges, 61% cross-team communication challenges, 75% AI usage, 62% AI guidelines). • UK Government Digital Service — GOV.UK Content Principles: Conventions and Research Background (qualitative guidance on shared writing conventions).
Public reference collection (`SOURCES.md`, `PUB-01` to `PUB-09`, `reference_only`)	• **W3C Web Accessibility Initiative (WAI / WCAG 2.2):** Informative Images (PUB-01), Decorative Images (PUB-02), Understanding Contrast Minimum (PUB-03), Understanding Use of Color (PUB-04). • **Public style & brand guideline examples:** UK Home Office Voice and Tone (PUB-05), U.S. Digital.gov Style Guide — Plain Language & Headlines (PUB-06, PUB-07), Chicago Bears Brand Guidelines — Clear Space & Minimum Logo Size (PUB-08, PUB-09).
Technical stack documentation (`PRESENTATION.md`)	• **LLM & Retrieval Models:** Sarvam API Model Catalogue, Sarvam Pricing & Credits, Hugging Face `intfloat/multilingual-e5-small`, Hugging Face `BAAI/bge-reranker-v2-m3`. • **Database, Storage & Hosting:** pgvector Documentation, Render Supported PostgreSQL Extensions, Render ParadeDB Deployment Guide, Cloudflare R2 Documentation, Vercel Documentation, Render Documentation.